

The WIN Time Theorem: Internal Derivation of Time and Its Arrow from the Five Layer Loop

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September 23, 2026

Abstract

We derive time as an internal structure of the WIN Paradigm. No external time coordinate is assumed. The five layers

$$\text{FDLM/WIN} \rightarrow \text{DME} \rightarrow \text{Core} \rightarrow \text{EFT} \rightarrow \text{Processor } T$$

compose into the diagonal loop operator $M_{\text{loop}} = T L_{\text{EFT}} L_{\text{Core}} L_{\text{DME}} L_{\text{FDLM}}$ acting on $\mathbb{C}^{14}$. On each observable mode m , the loop operator has the polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$. The *complex time* of mode m is

$$\tau_m \equiv \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m),$$

where ω_m is the mode's winding-generator frequency, $T_{\text{DME}} = 1/(2\gamma)$ is the DME relaxation time, and r_m is the mode's per-pass amplitude. The real part θ_m is the clock phase; the imaginary part $-\ln r_m > 0$ is the arrow increment. The processor T is exactly the projection onto modes with $\text{Re}(\tau_m) > 0$. Time is simultaneously the measurement outcome and the selection criterion of the loop: the loop observes itself by reading its own complex time. We state the WIN Time Theorem, give the level-by-level derivation, and identify what is derived, assumed, and open. Three algebraic bridges are closed: (i) the Core perturbation floor function is derived from a topological winding condition across the DME Core interface; (ii) the exact ratio $|M_A|/|M_B| = 3$ is derived by exact algebraic substitution without decimal approximation; (iii) dimensional homogeneity of τ_m is proved by showing θ_m and $-\ln r_m$ share the same scaling dimension along the Core flow.

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1 Introduction

1.1 The problem

The Standard Model has no fundamental arrow of time: time is an external parameter, and its direction is accounted for by statistical mechanics. The WIN Paradigm supplies a closed five layer loop with a preferred direction and a dissipative operator. In this paper we show that time is not an input to WIN; it is a readout of the loop.

Two distinct structures are separated:

1. **Clock time** – the phase accumulated by M_{loop} per pass.
2. **Arrow time** - the directed decay of M_{loop}^k because $|M_j| < 1$.

The first gives clocks. The second gives the arrow. Both arise from one operator.

1.2 What is assumed

The paper takes the following as given from the WIN Paradigm:

- the 8×8 torus substrate and the 14-mode multiplet at $\lambda = 4$;
- the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$;
- the DME stability coefficient $\gamma = 0.702839467990\dots$ and relaxation time $T_{\text{DME}} = 1/(2\gamma) = 0.711400003516965862\dots$;
- the transition rules $1 \rightarrow 2$ and $2 \rightarrow 3$;
- the EFT damping coefficient $\zeta H/\rho$ and the processor T .

It does not assume an external time coordinate, a Lorentzian metric, or a Wick rotation.

1.3 Organization

Section 2 states the five layer architecture. Sections 3-7 develop each level and identify the role time plays there. Section 8 defines the complex time operator and proves dimensional homogeneity. Section 9 proves the processor identity $T = \Pi_{\text{Re}(\tau) > 0}$. Section 10 derives the exact ratio $|M_A|/|M_B| = 3$. Section 11 states the WIN Time Theorem. Section 12 catalogs what is derived, assumed, and open.

2 The Five Layers and Their State Spaces

2.1 State spaces

#	Layer	State space	Operator type
1	FDLM/WIN	$\mathbb{C}^{14}$ mode vectors	Hermitian (unitary evolution)
2	DME	$\varepsilon \in \mathbb{R}$	scalar constraint
3	Core	$\varepsilon(\lambda)$ trajectory	gradient flow
4	EFT	field configurations	damped wave equation
5	Processor T	$\mathbb{C}^{14}$ projected	orthogonal projection

2.2 Loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}.$$

On the mode basis, M_{loop} is diagonal:

$$M_{\text{loop}}(m) = r_m e^{i\theta_m}.$$

The five layers are not temporally sequential within a pass. They are algebraically composed. Time in the loop is read from the eigenvalues of the composition, not from the duration of individual layers.

3 Level 1 — FDLM/WIN: Winding Phase

3.1 Substrate

The periodic lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has $N = 64$ sites and five-point Laplacian

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right).$$

The eigenvalue $\lambda = 4$ has multiplicity 14.

3.2 Winding generator

Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes and set

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}, \quad \phi = \frac{\pi}{8} M, \quad H_{\text{wind}} = i \sin \phi.$$

H_{wind} is Hermitian with spectrum

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}.$$

The beat period is $T_{\text{beat}} = 2\pi/(2\sqrt{2})$. At $t = T_{\text{beat}}$, $e^{iH_{\text{wind}}T_{\text{beat}}} = I$.

3.3 What time is doing at Level 1

Time at Level 1 is the *phase of the winding generator*. The eigenvalues of H_{wind} are the mode frequencies

$$\omega_m \in \{2\sqrt{2}, 6\sqrt{2}, 0\}.$$

The level is unitary; no amplitude is lost. It is a pure clock with an internal period T_{beat} .

The frequencies ω_m come in two non-trivial classes:

- class A: $\omega_A = 2\sqrt{2}$, three modes $\{\nu, e, t\}$;
- class B: $\omega_B = 6\sqrt{2}$, seven modes $\{u, d, s, \mu, c, \tau, b\}$;
- kernel: $\omega = 0$, four modes $\{\text{dark}, W, Z, H\}$.

The ratio $\omega_B/\omega_A = 3$ is exact.

4 Level 2 — DME: Time Quantum and Topological Quantization

4.1 The stability theorem

DME fixes the vacuum dimension $D = 4$ as a strict local maximum of the vacuum entropy. The stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990072464\dots,$$

where $I(2) = 0.400929882632204082\dots$ is the Ryu–Takayanagi strip integral at $\alpha = 2$ and $c_2 = 0.281789345410541014\dots$ is the framework’s DME constant. The second derivative of the entropy well at $\varepsilon \equiv D - 4 = 0$ satisfies

$$\frac{S''(0)}{S_0} = -2\gamma < 0,$$

so $D = 4$ is a strict local maximum.

4.2 The DME time quantum

The relaxation time is

$$T_{\text{DME}} = \frac{1}{2\gamma} = 0.711400003516965862\dots$$

4.3 Transition $1 \rightarrow 2$: the phase handoff

The unique dimensionless quantity at the $1 \rightarrow 2$ boundary is

$$\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}}.$$

For class A, $\Phi_A = 2 + \delta_{\text{edge}}$; for class B, $\Phi_B = 6 + 3\delta_{\text{edge}}$, where

$$\delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2 = 0.012143066491921761\dots$$

4.4 Topological derivation of the floor function

A referee might object that the transition rule $\varepsilon_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$ is a postulated integer-floor rule. We derive it from a winding condition on the DME-Core interface.

Definition 1 (DME-Core interface). *The DME-Core interface is a $U(1)$ -valued map from the loop parameter space S^1 (one loop pass) to the DME phase space, given by*

$$\gamma_m : S^1 \rightarrow U(1), \quad \gamma_m(s) = e^{2\pi i \omega_m T_{\text{DME}} s}, \quad s \in [0, 1).$$

Lemma 1 (Lift and winding). *The lift of γ_m to $\mathbb{R}$ is the monotonically increasing function $\tilde{\gamma}_m(s) = \omega_m T_{\text{DME}} s$. Over one loop pass ($s : 0 \rightarrow 1$), the lift crosses integer values $k \in \mathbb{Z}$ exactly $\lfloor \omega_m T_{\text{DME}} \rfloor$ times (the final point excluded).*

Proof. The lift $\tilde{\gamma}_m$ ranges over $[0, \omega_m T_{\text{DME}})$. The integers in this half-open interval are $0, 1, \dots, \lfloor \omega_m T_{\text{DME}} \rfloor - 1$, counted with multiplicity 1 each. The remaining integer $\lfloor \omega_m T_{\text{DME}} \rfloor$ is the ceiling reached at the endpoint $s = 1$. Therefore the number of complete DME cycles per pass is $n_m \equiv \lfloor \omega_m T_{\text{DME}} \rfloor$. $\square$

Definition 2 (DME natural period). *The DME layer has a natural phase period of 1 in the loop time unit T_{DME} . That is, one complete DME cycle corresponds to advancing the loop parameter by one unit of T_{DME} .*

The natural period 1 in loop units is not imposed from outside; it is the definition of the DME relaxation cycle. Because the DME stability coefficient γ sets a single relaxation rate 2γ , one relaxation time $T_{\text{DME}} = 1/(2\gamma)$ is the natural DME cycle.

4.5 Transition 2 $\rightarrow$ 3: cycle-quantized Core perturbation

Definition 3 (DME-Core boundary functional). *Define $J[\varepsilon_0] = \varepsilon_0 + \Lambda \cdot \mathbf{1}[\varepsilon_0 \leq |w+1|]$ with $\Lambda \rightarrow \infty$.*

Theorem 1 (Core perturbation quantization). *The unique minimizing ε_0 of J on the lattice $\delta_{\text{edge}}\mathbb{Z}_{>0}$, subject to the winding condition of Lemma 1, is*

$$\varepsilon_0(m) = n_m \cdot \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor.$$

Proof. The functional J is finite only if $\varepsilon_0 > |w+1|$. The lattice is $\delta_{\text{edge}}\mathbb{Z}_{>0}$. The winding condition fixes $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$. We must verify that $n_m \cdot \delta_{\text{edge}} > |w+1|$ for both non-trivial classes.

Numerically: $|w+1| = 0.014054078857664869$ and $\delta_{\text{edge}} = 0.012143066491921761$, so $|w+1|/\delta_{\text{edge}} = 1.1573747757\dots$. Thus any $n \geq 2$ satisfies $n \cdot \delta_{\text{edge}} > |w+1|$.

For class A: $n_A = 2$, so $\varepsilon_0(A) = 2\delta_{\text{edge}} = 0.024286132983843522 > |w+1|$. Satisfied.

For class B: $n_B = 6$, so $\varepsilon_0(B) = 6\delta_{\text{edge}} = 0.072858398951530566 > |w+1|$. Satisfied.

Thus the winding-quantized ε_0 automatically satisfies the boundary constraint for both classes, and the minimizing element on the lattice is unique. $\square$

Remark 1. *The floor function is therefore not an ansatz. It is the winding number of the DME phase over one loop pass, computed on the $U(1)$ -valued interface. The boundary constraint $\varepsilon_0 > |w+1|$ is a consistency condition automatically satisfied by the winding-quantized value.*

4.6 What time is doing at Level 2

Level 2 has no internal dynamics. It *imposes the time quantum T_{DME} at the $1 \rightarrow 2$ boundary and quantizes the Core perturbation* by the DME phase winding number. This is the only level where the time unit crosses a transition. The output to Core is

$$\{T_{\text{DME}}, n_m = \lfloor \omega_m T_{\text{DME}} \rfloor, \varepsilon_0(m) = n_m \delta_{\text{edge}}, S''(0)/S_0 = -2\gamma\}.$$

5 Level 3 — Core: The Arrow of Time

5.1 Flow equation

Core obeys

$$\frac{d\varepsilon}{d\lambda} = -2\gamma\varepsilon.$$

The solution is $\varepsilon(\lambda) = \varepsilon_0 e^{-2\gamma\lambda}$.

5.2 Lyapunov functional

$$L[\varepsilon] = \gamma\varepsilon^2, \quad \frac{dL}{d\lambda} = -4\gamma^2\varepsilon^2 \leq 0,$$

strict for $\varepsilon \neq 0$. The Core flow is strictly monotone.

5.3 Master conditions

M1 (monotonicity): $dL/d\lambda \leq 0$. Satisfied. M2 (irreversibility): under $\lambda \rightarrow -\lambda$, L increases. Satisfied. No oscillation: characteristic equation $s + 2\gamma = 0$ has real root $s = -2\gamma$.

5.4 Transition $3 \rightarrow 4$

The Core-to-EFT transition occurs when $\varepsilon(\lambda_{\text{EFT}}) = |w + 1|$:

$$\lambda_{\text{EFT}}(m) = \frac{\ln(\varepsilon_0(m)/|w + 1|)}{2\gamma}.$$

For class A: $\lambda_{\text{EFT}} = 0.389130725768564\dots$ For class B: $\lambda_{\text{EFT}} = 1.170683511790840\dots$

5.5 What time is doing at Level 3

Time at Level 3 is the *flow parameter* λ . The state decays monotonically. The Lyapunov functional decreases. This is the source of the arrow: the direction of increasing λ is the direction of decreasing L . No statistical assumption is used.

6 Level 4 — EFT: Damping

6.1 Mode equation

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2 \right) \phi = 0,$$

with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \xi R$.

6.2 EFT parameters

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi}, \quad \frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi}.$$

The EFT decay factor per beat period is

$$D_{\text{factor}} = \exp\left(-\frac{\zeta H}{\rho} \frac{T_{\text{beat}}}{2}\right) = 0.994810128563526019\dots$$

6.3 Regime

$$\frac{\zeta R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}.$$

The EFT is extremely underdamped, mass-dominated, gradient-negligible. All 14 modes oscillate at the same frequency m_s and decay at the same rate $\Gamma_{\text{eff}}/2$.

6.4 What time is doing at Level 4

Time at Level 4 is the *damped-oscillation time*. The field oscillates at ω_m and damps at $\Gamma_{\text{eff}}/2$. The EFT is a coarse-grained description; it does not resolve individual loop passes.

7 Level 5 — Processor T

7.1 Definition

$$T = P_{\text{mesh}} + P_{\tau},$$

where P_{mesh} projects onto the nine mesh-resonance modes $\{\nu, e, u, d, s, \mu, c, b, t\}$ and P_{τ} projects onto the rank-1 mode τ .

7.2 Properties

$T^2 = T$, $T^\dagger = T$, $\text{rank } T = 10$. Kernel: $\{\text{dark}, W, Z, H\}$.

7.3 Charge bookkeeping

$$\sum_{j \in \text{image}(T)} Q_j = -2, \quad \sum_{j \in \text{kernel}(T)} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1.$$

7.4 What time is doing at Level 5

Time at Level 5 is the *selection criterion*. The processor reads the complex time of each mode and selects those with $\text{Re}(\tau_m) > 0$. Modes with $\tau_m = 0$ are killed. The kernel modes are frozen under M_{loop} .

8 The Complex Time Operator

Definition 4 (Complex time). *For each mode m in the observable sector, the complex time is*

$$\tau_m \equiv \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m).$$

The real part $\text{Re}(\tau_m) = \omega_m T_{\text{DME}}$ is the clock phase per pass; the imaginary part $-\text{Im}(\tau_m) = -\ln r_m > 0$ is the arrow increment per pass.

Proposition 1 (Closed form of τ_m). *For the two nontrivial classes,*

$$\tau_A = 2.0121430664919213 - 0.5521962501528914 i,$$

$$\tau_B = 6.0364291994757650 - 1.6508085388210012 i.$$

The kernel has $\tau_C = 0$.

Proof. Direct from $\omega_A = 2\sqrt{2}$, $\omega_B = 6\sqrt{2}$, $T_{\text{DME}} = 1/(2\gamma)$, and r_A, r_B computed from the loop-decay decomposition. $\square$

8.1 Loop-decay decomposition

The loop amplitude decomposes exactly as

$$r_m = \underbrace{\frac{|w+1|}{\varepsilon_0(m)}}_{\text{Core factor}} \cdot \underbrace{D_{\text{factor}}}_{\text{EFT factor}}.$$

Numerically:

$$\begin{aligned} r_A &= 0.5786873879 \times 0.9948101286 = 0.575684074716075633, \\ r_B &= 0.1928957960 \times 0.9948101286 = 0.191894691572025183. \end{aligned}$$

The Core and EFT contributions compose multiplicatively. They are distinct levels' contributions to a single per-pass amplitude reduction.

8.2 Dimensional homogeneity of τ_m

A referee might object that the additive combination of θ_m and $\ln r_m$ inside τ_m requires proof that the two quantities share the same scaling dimension. We provide it here.

Proposition 2 (Dimensional homogeneity). *Both θ_m and $-\ln r_m$ are per-pass quantities of the form $(\text{rate}) \times T_{\text{DME}}$, where rate has dimensions of inverse time. Specifically:*

$$\theta_m = \omega_m T_{\text{DME}}, \quad -\ln r_m = \frac{\Gamma_m}{2} T_{\text{DME}},$$

where $\Gamma_m = -2 \ln r_m / T_{\text{DME}} > 0$ is the mode's per-pass decay rate. Under the Core flow parameter λ , both quantities scale linearly with λ .

Proof. The phase accumulated over one loop pass is $\theta_m = \omega_m T_{\text{DME}}$, which is the product of the mode frequency ω_m (units: inverse time) and the loop time unit T_{DME} (units: time).

The log-amplitude decrement per pass is $-\ln r_m$. Define $\Gamma_m \equiv -2 \ln r_m / T_{\text{DME}}$. Then $-\ln r_m = (\Gamma_m/2) T_{\text{DME}}$, which is the product of the decay rate $\Gamma_m/2$ (units: inverse time) and T_{DME} .

Both θ_m and $-\ln r_m$ therefore have the form $(\text{rate}) \times T_{\text{DME}}$. Both are dimensionless (rate $\times$ time). Both are linear in the continuous time coordinate $t = k T_{\text{DME}}$, since after k passes $\theta_m \cdot k = \omega_m t$ and $-\ln r_m \cdot k = (\Gamma_m/2) t$.

Under the Core flow parameter λ , the Core amplitude decays as $\varepsilon(\lambda) = \varepsilon_0 e^{-2\gamma\lambda}$, so the log-amplitude $\ln \varepsilon(\lambda) = \ln \varepsilon_0 - 2\gamma\lambda$ is linear in λ . The per-pass log decrement $-\ln r_m$ is therefore proportional to the Core flow parameter, and the phase θ_m is likewise linear in the loop parameter. The two quantities share the same scaling dimension and the same affine dependence on the flow parameter.

Therefore the additive combination $\tau_m = \theta_m + i \ln r_m$ is dimensionally homogeneous and $\ln r_m$ transforms as an authentic temporal increment under the layer transitions. $\square$

8.3 Structural identities

$$\frac{\theta_B}{\theta_A} = 3 \quad (\text{exact}), \quad \ln \frac{r_A}{r_B} = \ln 3 \quad (\text{exact}).$$

9 The Processor as Projection onto Nonzero Time

Theorem 2 (Processor = projection onto nonzero real time). *Let $T = P_{\text{mesh}} + P_\tau$ be the processor. Then*

$$T = \Pi_{\text{Re}(\tau) > 0},$$

i.e. T keeps exactly those modes with $\text{Re}(\tau_m) > 0$ and kills those with $\tau_m = 0$.

Proof. By direct computation on the 14-mode multiplet: every mode in the image of T (that is, every mode with $\delta \neq 0$ or $s = +1$) has $\omega_m \in \{2\sqrt{2}, 6\sqrt{2}\}$, hence $\text{Re}(\tau_m) = \omega_m T_{\text{DME}} > 0$. Every mode in the kernel of T (that is, every mode with $\delta = 0$ and $s \neq +1$) has $\omega_m = 0$, hence $\tau_m = 0$. $\square$

Remark 2. *The image of T is exactly the set of modes for which the loop produces nonzero complex time. The kernel of T is exactly the set of modes for which the loop produces no time.*

10 Exact Ratio $|M_A|/|M_B| = 3$

Theorem 3 (Exact ratio). *The ratio of the two nontrivial loop-eigenvalue magnitudes is exactly 3:*

$$\frac{|M_A|}{|M_B|} = \frac{r_A}{r_B} = 3.$$

The framework constants $|w + 1|$, δ_{edge} , and D_{factor} do not appear in the ratio.

Proof. From the loop-decay decomposition,

$$r_m = \frac{|w + 1|}{\varepsilon_0(m)} D_{\text{factor}}, \quad \varepsilon_0(m) = n_m \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor.$$

Therefore

$$\frac{r_A}{r_B} = \frac{|w + 1|/(n_A \delta_{\text{edge}}) \cdot D_{\text{factor}}}{|w + 1|/(n_B \delta_{\text{edge}}) \cdot D_{\text{factor}}} = \frac{n_B}{n_A}.$$

We now compute n_A, n_B exactly. By the handoff,

$$\omega_A T_{\text{DME}} = \frac{2\sqrt{2}}{2\gamma} = \frac{\sqrt{2}}{\gamma}, \quad \omega_B T_{\text{DME}} = \frac{6\sqrt{2}}{2\gamma} = \frac{3\sqrt{2}}{\gamma}.$$

With $\gamma = 0.702839467990072464\dots$,

$$\frac{\sqrt{2}}{\gamma} = 2.0121430664919213\dots, \quad \frac{3\sqrt{2}}{\gamma} = 6.0364291994757650\dots$$

Write $\sqrt{2}/\gamma = 2 + \delta_{\text{edge}}$ where $\delta_{\text{edge}} = 0.012143066491921761\dots$. Then $3\sqrt{2}/\gamma = 6 + 3\delta_{\text{edge}}$ with $3\delta_{\text{edge}} = 0.036429199475765284\dots < 1$. Therefore

$$n_A = \lfloor 2 + \delta_{\text{edge}} \rfloor = 2, \quad n_B = \lfloor 6 + 3\delta_{\text{edge}} \rfloor = 6.$$

Hence

$$\frac{r_A}{r_B} = \frac{n_B}{n_A} = \frac{6}{2} = 3.$$

The framework constants $|w + 1|$ and D_{factor} cancel identically; the δ_{edge} factor cancels between numerator and denominator because both appear in the form $n_m \delta_{\text{edge}}$. The ratio is exact and independent of decimal approximation. $\square$

Corollary 1. *Since $|M_{\text{loop}}(m)| = r_m$ for observable m , we have*

$$\frac{|M_A|}{|M_B|} = \frac{r_A}{r_B} = 3.$$

11 The WIN Time Theorem

Theorem 4 (WIN Time Theorem). *In the five-layer WIN Paradigm:*

1. No external time coordinate is assumed. The loop operator M_{loop} is the only time-evolution object.
2. On each observable mode m , $M_{\text{loop}}(m) = r_m e^{i\theta_m}$ with $\theta_m = \omega_m T_{\text{DME}}$ and $T_{\text{DME}} = 1/(2\gamma)$.
3. The complex time per pass is $\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$. Both components share the scaling dimension (rate) $\times T_{\text{DME}}$ (Proposition 2).
4. Discrete WIN time is the loop count $k \in \mathbb{Z}_{\geq 0}$. Continuous WIN time is

$$t = k T_{\text{DME}}.$$

Under this identification, the loop evolution is

$$|\psi_m(t)\rangle = e^{i\omega_m t} e^{-\Gamma_m t/2} |e_m\rangle, \quad \Gamma_m = \frac{-2 \ln r_m}{T_{\text{DME}}} > 0.$$

5. The arrow of time is the direction of increasing k , equivalently the direction of decreasing Core Lyapunov functional

$$L[\varepsilon] = \gamma \varepsilon^2, \quad \frac{dL}{d\lambda} = -4\gamma^2 \varepsilon^2 \leq 0,$$

equivalently the direction of increasing $-\ln r_m$.

6. The processor T is exactly the projection onto modes with nonzero real time:

$$T = \Pi_{\text{Re}(\tau) > 0}.$$

7. Kernel modes have $\tau_m = 0$ and are frozen under M_{loop} .

8. Exact ratio:

$$\frac{|M_A|}{|M_B|} = 3.$$

9. The floor function in the Core perturbation $\varepsilon_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \delta_{\text{edge}}$ is the winding number of the DME-Core interface phase over one loop pass (Lemma 1, Theorem 1).

12 What Is Derived, Assumed, and Open

12.1 Derived

- Loop operator diagonalization and complex time formula (Definition 4, Proposition 1).
- Closed form of τ_A, τ_B, τ_C .
- Multiplicative decomposition of the loop decay into Core and EFT factors.
- Topological derivation of the floor function from the DME-Core winding condition (Lemma 1, Theorem 1).

- Exact algebraic ratio $|M_A|/|M_B| = 3$ (Theorem 3).
- Dimensional homogeneity of τ_m (Proposition 2).
- Processor identity $T = \Pi_{\text{Re}(\tau)>0}$ (Theorem 2).
- Arrow of time from Core Lyapunov and loop dissipativity.
- Composition order forced by state-space chain.

12.2 Assumed

- The transition rule $1 \rightarrow 2$ uses T_{DME} as the phase unit.
- The DME natural phase period is 1 in loop time units.
- The continuous-time identification $t = kT_{\text{DME}}$.

12.3 Open

- **Loop pass duration.** Both T_{DME} (DME relaxation) and T_{beat} (Level-1 internal period) are dimensionless time-like quantities. The transition rule uses T_{DME} ; Level 1 uses T_{beat} internally. The framework identifies the loop pass duration with T_{DME} by the transition rule, but does not derive from a deeper principle that T_{DME} and not T_{beat} is the physical pass duration. Both scales coexist in the framework; their ratio $T_{\text{beat}}/T_{\text{DME}} = 3.1226\dots$ is a structural constant.
- **Lorentzian signature.** The Hodge complex on the 8×8 torus is Euclidean. The framework does not derive the Lorentzian signature $(1, 3)$ from first principles. This is an open problem shared with all discrete approaches to quantum gravity.
- **Physical conversion.** The framework measures time in units of T_{DME} . Conversion to seconds would require choosing a physical process to calibrate against. This choice is external to WIN.

13 Conclusion

Time in the WIN Paradigm is not an external coordinate. It is the complex eigenvalue of the loop operator,

$$\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m).$$

Its real part is the clock. Its imaginary part is the arrow. The processor T is the projection onto nonzero real time. The loop observes itself by reading its own complex time. No external time unit is required: one pass through the loop advances WIN time by one unit of T_{DME} .

Three algebraic bridges have been closed in this paper: the floor function of the Core perturbation is the winding number of the DME-Core interface (Lemma 1, Theorem 1); the ratio $|M_A|/|M_B| = 3$ is exact by algebraic cancellation of framework constants (Theorem 3); and dimensional homogeneity of the complex time follows because both θ_m and $-\ln r_m$ are per-pass products of a rate with T_{DME} (Proposition 2).

The framework derives time and its arrow from first principles, using only WIN-native objects. The three open items (loop pass duration selection, Lorentzian signature, physical conversion) are stated explicitly and are not required for the internal derivation.

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A Bibliographic Metadata for AI Indexing

This appendix provides structured bibliographic metadata in multiple formats for machine indexing and retrieval-augmented generation systems.

Dublin Core Metadata

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DC.title: The WIN Time Theorem: Internal Derivation of Time and Its Arrow from the Five-Layer
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09-23
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.subject: WIN Paradigm; FDLM; Hodge Complex; DME; ENTROPIA-Core; ENTROPIA-EFT; Complex Time;
DC.description: We derive time as an internal structure of the WIN Paradigm. No external time
DC.identifier: WIN-PARADIGM-TIME-2026-09-23
DC.publisher: Entropia Research Institute / Information Physics Institute
DC.rights: All rights reserved
DC.relation.isPartOf: WIN Paradigm White Paper Series
DC.relation.references: Preschutti Motion Theorem (2026); Preschutti WIN Paradigm v2 (2026); P

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Schema.org JSON-LD

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    "Emergent Time",
    "Processor Projection"
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  "publisher": {
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    "name": "Entropia Research Institute"
  },
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  "identifier": "WIN-PARADIGM-TIME-2026-09-23"
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BibTeX

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  journal     = {WIN Paradigm White Paper Series},
  year        = {2026},
  month       = {September},
  day         = {23},
  note        = {ORCID: 0009-0004-5445-1744},
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                 Arrow of Time, Loop Operator, 8x8 Torus,
                 Quantum Foundations, Emergent Time},
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                 Paradigm. No external time coordinate is assumed.
                 The five layers FDLM/WIN -> DME -> Core -> EFT ->
                 Processor compose into a diagonal loop operator on
                  $C^{14}$ . The complex time  $\tau_m = \theta_m + i \ln r_m$ 
                 is the eigenvalue the loop reads. The processor T
                 is exactly the projection onto modes with
                  $\text{Re}(\tau) > 0$ .}
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  year        = {2026},
  month       = {September},
  howpublished = {WIN Paradigm White Paper}
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  title       = {{WIN} Paradigm --- Technical Handoff Document,
                 Version 3, After the Motion Theorem},
  year        = {2026},
  month       = {September},
  howpublished = {Working Technical Document}
}

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                 {ENTROPIA-Core}, {ENTROPIA-EFT}, and {ENTROPIX/MESA}
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year      = {2026},
month     = {September},
howpublished = {WIN Paradigm White Paper}
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AI-Crawler Semantic Summary

TITLE: The WIN Time Theorem
 AUTHOR: Stanley Preschutti (ORCID 0009-0004-5445-1744)
 DATE: 2026-09-23
 SERIES: WIN Paradigm White Paper

CORE CLAIM:

Time is derived as the complex eigenvalue of the WIN loop operator $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ acting on C^{14} . No external time coordinate is assumed. The complex time of mode m is $\tau_m = \omega_m T_{\text{DME}} + i \ln r_m$. The real part is the clock; the imaginary part is the arrow. The processor T equals the projection onto modes with $\text{Re}(\tau_m) > 0$.

KEY EQUATIONS:

```

tau_m = omega_m T_DME - i(-ln r_m)
T_DME = 1/(2 gamma) = 0.711400003516965862
omega_A = 2 sqrt(2); omega_B = 6 sqrt(2)
tau_A = 2.0121430664919213 - 0.5521962501528914 i
tau_B = 6.0364291994757650 - 1.6508085388210012 i
|M_A| / |M_B| = 3 (exact)
epsilon_0(m) = floor(omega_m T_DME) * delta_edge
r_m = (|w+1| / epsilon_0(m)) * D_factor
T = Pi_{Re(tau) > 0}

```

RESULTS ESTABLISHED:

1. Time = complex eigenvalue of the loop operator.
2. Clock = real part; arrow = imaginary part.
3. Processor = projection onto nonzero real time.
4. Floor function derived from DME-Core interface winding.
5. Ratio $|M_A|/|M_B| = 3$ exact by algebraic cancellation.
6. Dimensional homogeneity of τ_m proven.

OPEN ITEMS:

1. Loop pass duration selection between T_{DME} and T_{beat} .
2. Lorentzian signature not derived (Euclidean Hodge complex).
3. Physical conversion to seconds requires external calibration.

PRECEDING PAPERS:

- Preschutti, WIN Motion Theorem (2026)
- Preschutti, WIN Paradigm v2 (2026)

- Preschutti, Technical Handoff v3 (2026)

CITATION KEY: PreschuttiWINTime2026